MODULE 02 · DOMAIN 1 - PLAN AND MANAGE AN AZURE AI SOLUTION

Choosing a model

LLMs, small models, reasoning and multimodal models, Foundry Tools, the catalogue, leaderboards and model router.

DOMAIN 1 - Plan and manage WEIGHT 25-30% MODULE 02 of 39

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Start from the task's **constraints**, not the leaderboard. The right model is the smallest one that passes your evaluation.

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Kinds of model, and what each is for

KIND	STRONG AT	REACH FOR IT WHEN
Large language model	Broad knowledge, open-ended language	The task needs judgement or flexibility
Small language model	Focused tasks, fast and cheap	The task is narrow, or must run locally
Reasoning model	Planning, multistep analysis	Accuracy on hard problems beats speed
Multimodal model	Text with images, and sometimes audio	The input is not only text
Embedding model	Turning text into vectors	You need vector search, not generation
Foundry Tool	One job, done predictably	A purpose-built service already exists

ON THE EXAM

Embedding models never write answers. If the requirement is similarity search, the answer is an embedding model, not a chat model.

Large or small language model?

LARGE LANGUAGE MODEL

Breadth

- Hundreds of billions of parameters
- Strong across varied, open tasks
- Higher cost and latency per call
- Runs in the cloud

SMALL LANGUAGE MODEL

Efficiency

- Typically under 1 to about 14 billion parameters
- Can match larger models on focused tasks
- Runs on a single GPU, or even CPU
- Can run on premises or on a device

THE TIEBREAKER

Offline, on a device, or no data leaving the machine: that is a small model's scenario. Phi is Microsoft's small model family.

Foundry Local: models on the user's device

- Runs models **on the user's own hardware**, so the app works **offline**.
- No Azure subscription needed at runtime. Uses the GPU or NPU if present, otherwise CPU.
- **OpenAI-compatible API**, so existing OpenAI SDK code can point at it.
- Curated catalogue of small, compressed models, including Phi and Whisper for transcription.

> ON THE EXAM
>
> Foundry Local answers 'no connectivity' and 'nothing leaves the device'. It does not give you cloud-scale models.

Reasoning models and reasoning effort

- Reasoning models work through a problem **before** answering. Better on planning and hard analysis.
- **reasoning_effort** controls how much. Lower effort means faster responses and fewer reasoning tokens.
- Reasoning tokens cost money and add latency, even though users never see them.
- Defaults and allowed values **differ between model versions**. Swapping a deployment is not a drop-in change.

> THE TIEBREAKER
>
> Live chat answering simple questions: low or no reasoning. Planning a multistep agent task: more reasoning.

Multimodal and generation models

- **Multimodal models** reason over text and images together: describe, compare, answer questions about a picture.
- Some models take **audio** input directly. Speech modules cover when that beats transcription first.
- **Generation models** create media: gpt-image models for images, Sora for video.
- Understanding an image and generating one are different models. Do not mix them up.

> ON THE EXAM
>
> Domain 3 goes deep on all of this. For model choice, just match the model to the input and output types.

A Foundry Tool, or a generative model?

REQUIREMENT	BETTER FIT
Detect and redact PII consistently, at high volume	Azure Language PII detection
Transcribe recorded calls to text	Azure Speech
Translate whole documents, keeping their formatting	Azure Translator document translation
Pull fields from invoices into a fixed schema	Content Understanding
Summarise the themes in customer feedback	A language model
Answer open questions about company policy	A language model, grounded with retrieval

THE TIEBREAKER

Narrow task, same output every time: a Foundry Tool. Judgement or open-ended language: a model.

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Two kinds of Foundry Model

SOLD BY AZURE

Microsoft stands behind it

- All Azure OpenAI models, plus selected others
- Billed through your Azure subscription
- Azure SLAs and Microsoft support
- Some share fungible provisioned throughput

FROM PARTNERS AND COMMUNITY

The provider stands behind it

- Most of the catalogue, from many providers
- Support and SLAs vary by provider
- Some require Azure Marketplace
- Deployment options chosen by the provider

THE SIGNAL

A requirement for enterprise SLAs and Microsoft support points to a model sold by Azure.

Reading the leaderboards correctly

- **Quality index:** higher is better, across reasoning, coding, maths and knowledge.
- **Safety:** measured as attack success rate, so **lower** is better.
- **Cost:** an estimate assuming three input tokens for every output token.
- **Throughput and latency:** how responsive the model is in practice.
- **Trade-off charts** plot two metrics; side-by-side comparison takes up to three models.

> ON THE EXAM
>
> If your workload is output-heavy, the cost leaderboard's three-to-one assumption understates your real cost.

A benchmark tells you what to **shortlist**. Only an evaluation on your own data tells you what to **choose**.

A model selection process

01
Constraints
Offline? Region? Modality? Latency? Budget?

02
Shortlist
Filter the catalogue; use leaderboards.

03
Compare
Trade-off charts and side-by-side views.

04
Evaluate
Test the shortlist on your own data.

05
Deploy
The smallest model that passes.

06
Re-check
Evaluate again when models change.

ON THE EXAM

Most wrong answers skip a step: usually constraints at the start, or evaluation in the middle.

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Model router

- A **trained model** that sends each prompt to a suitable underlying model, behind **one deployment**.
- **Balanced**, the default: among models close to the best quality, pick the cheapest.
- **Cost:** accepts a wider quality gap to save more. **Quality:** best model, cost ignored.
- **Model subset** limits which models can be used. Failover to another model is on by default.

> ON THE EXAM
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> Mixed traffic of simple and hard prompts, with one endpoint to manage: model router. Still evaluate it against your baseline.

Where model router catches people out

- The **context window is the smallest model's**. Long prompts fail unless a subset raises that floor.
- It accepts images, but **routes on the text alone**. It does **not** process audio.
- New models are **not** added to a subset automatically. Failover needs **at least two** models in it.
- Claude models must be deployed first before the router can use them.
- Auto-update changes the model pool, which can change quality and cost.

> THE SIGNAL
>
> Intermittent 'context exceeded' errors on a router deployment point straight at model subset.

> An energy firm equips engineers with rugged laptops for offshore wind farms, where there is no network connection for days at a time.
>
> Engineers need to ask questions about maintenance manuals stored on the laptop. Policy forbids any manual content leaving the device.

Which approach should the firm use?

A. A large model on a Global Standard deployment, queuing questions until the connection returns.

B. A small language model running on the laptop through Foundry Local.

C. Model router in Cost mode, to keep the per-question cost low.

D. Azure Language summarisation of each manual, cached on the laptop.

◆ PAUSE AND DECIDE

Which approach should the firm use?

A. Large cloud model, queue until online.
Days of waiting fails the need, and sending manuals to the cloud breaks the policy.

B. Small model on the laptop via Foundry Local.
Runs offline on the device's own hardware, so nothing leaves it.

C. Model router in Cost mode.
Still a cloud deployment. Cost is not the constraint here; connectivity is.

D. Azure Language summarisation, cached.
Needs the cloud to create, cannot answer new questions, and the feature is retiring.

THE SIGNAL

Two constraints in two sentences: no network, and nothing leaves the device. Each one alone eliminates every cloud option.

> A legal team deploys model router with default settings to analyse contracts. Short questions about clauses work well.
>
> When users paste entire long contracts, some requests fail with a context-length error, while identical requests sometimes succeed.

What should the team change?

A Switch the router's routing mode from Balanced to Quality.

B Raise the tokens-per-minute rate limit on the router deployment to allow bigger requests.

C Enable auto-update so the router gains newer models.

D Configure a model subset of only models with large enough context windows.

◆ PAUSE AND DECIDE

What should the team change?

A Switch to Quality mode.

Quality ranks models by answer quality, not context size. A smaller-window model can still be chosen.

B Raise the TPM limit.

Rate limits govern volume over time. This error is about the size of one request.

C Enable auto-update.

It changes the pool, but the smallest model still sets the window, and quality may shift.

D Model subset of large-context models only.

Raises the smallest context window in the pool, which is the router's effective limit.

THE SIGNAL

'Identical requests sometimes succeed' means the outcome depends on which model was picked. That is a pool problem.

A bank stores two million customer emails a month for analysis. Before storage, it must redact names, account numbers and addresses.

Compliance requires consistent, documented detection categories that auditors can check, at the lowest reasonable cost.

Which approach best meets the requirements?

A Azure Language PII detection in Foundry Tools, redacting the detected entities.

B A large reasoning model prompted to find and redact all personal information.

C Model router in Cost mode, with a redaction prompt applied to each email.

D Fine-tune a small language model on thousands of manually redacted bank emails.

◆ PAUSE AND DECIDE

Which approach best meets the requirements?

A Azure Language PII detection.

Purpose-built, documented entity categories, predictable output, and a feature with continued support.

B Large reasoning model with a prompt.

Often accurate, but output can vary, categories are not documented, and reasoning tokens are costly at volume.

C Model router in Cost mode.

Cheaper, but different models may handle different emails, which undermines consistency.

D Fine-tune a small model.

Training and upkeep effort, with no documented categories and no reason to beat a purpose-built tool.

THE SIGNAL

'Consistent', 'documented' and 'auditors can check' are words for a purpose-built tool, not for a prompt.

> A retailer picked the model at the top of the quality leaderboard for a customer chatbot. It answers simple questions about order status and returns.
>
> Quality is fine, but monthly cost is three times the budget. The team must cut cost without harming answer quality.

What should the team do next?

A. Deploy whichever model tops the cost leaderboard, since quality is already acceptable.

B. Switch to whichever model has the lowest attack success rate on the safety leaderboard.

C. Shortlist cheaper models on a trade-off chart, then evaluate them on real conversations.

D. Keep the current model, but cap every response at 50 output tokens to cut spend.

◆ PAUSE AND DECIDE

What should the team do next?

A. Top of the cost leaderboard.

Repeats the original mistake from the other direction: choosing from a benchmark with no evaluation.

B. Lowest attack success rate.

Safety matters, but it is not the problem stated, and it does nothing for cost.

C. Shortlist on a trade-off chart, then evaluate on real conversations.

Shortlist from benchmarks, decide on evidence from the actual task.

D. Cap output at 50 tokens.

Truncates answers, harming quality, while the oversized model stays in place.

THE TIEBREAKER

Simple questions on an expensive frontier model is the classic over-sized choice. Fix the model, then prove it with evaluation.

Choosing a model in five lines

- **Constraints first:** offline, region, modality, latency and budget eliminate options.
- **Narrow and repeatable:** a Foundry Tool. **Judgement:** a model.
- **Small models** for focused or on-device work. **Reasoning** only where thinking pays.
- **Leaderboards shortlist; evaluation decides.** Safety scores: lower is better.
- **Model router** for mixed traffic. Its context window is its smallest model's.

> ON THE EXAM
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> Next, module 03: choosing the Foundry service around the model, for generation, grounding, search, agents and multimodal work.